CS516/CSL516: Parallelization of Programs

Dependency Analysis

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Recap

- Loop Dependency Analysis
  - Distance Vector
  - Direction Vectors

Exercise-1

```
    for (i = 1; i<=N; i++)
        for (j = 1; j<=M; j++)
S1:         A[i+1][j+1] = D[i][j] + 10
S2:         B[i][j] = A[i][j] + 20
```

Compute the direction vectors

Exercise-2

```
DO J = 1, 100
    DO I = 1, 100
        A(I+1) = A(I) + B(J)
    ENDDO
ENDDO
```

Compute the direction vectors

- The direction vectors for the dependencies is (*,<)

Exercise-3

```
DO I = 1, 100
    DO J = 1, 100
S1:     A(I+1, 5) = A(I, N)
    ENDDO
ENDDO
```

Compute the direction vectors

Exercise-4

```
DO I = 1, N
    DO J = 1, M
        DO K = 1, L
            A(I+1,J,K) = A(I,J,K) + X1
            B(I,J,K+1) = B(I,J,K) + X2
            C(I+1,J+1,K+1) = C(I,J,K) + X3
        ENDDO
    ENDDO
ENDDO
```

Compute the direction vectors

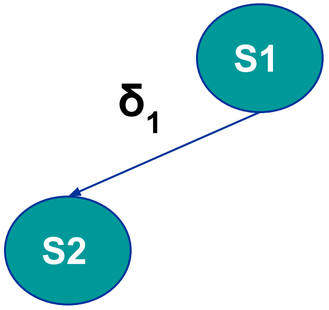
Exercise-5

```
    DO I = 1, N
S1      A(I+1) = B(I) + C
S2      D(I) = A(I) + E
    ENDDO
```

- Does it have any dependency?

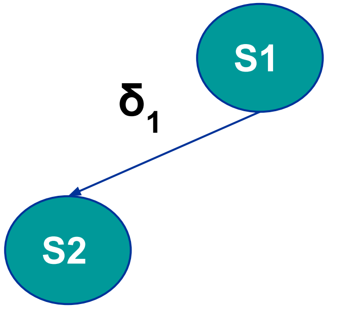
  Anti or True?

  - **S1 δ_1 S2**



Exercise-6

```
    DO I = 1, N
S1      A(I+1) = B(I) + C
S2      D(I) = A(I) + E
    ENDDO
```

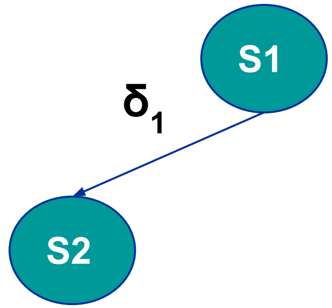


- Can we parallelize the loop?
- Can we transform transform (split) the loops and extract parallelism?

Exercise-6

```
    DO I = 1, N
S1      A(I+1) = B(I) + C
S2      D(I) = A(I) + E
    ENDDO
```

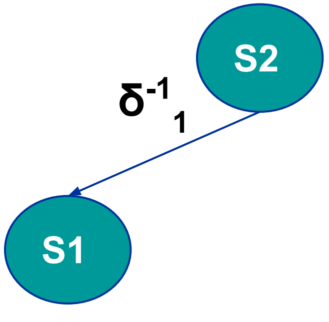
```
    PARALLEL DO I = 1, N
S1      A(I+1) = B(I) + C
    ENDDO

    PARALLEL DO I = 1, N
S2      D(I) = A(I) + E
    ENDDO
```



Exercise-7

```
    DO I = 1, N
S1       A(I) = B(I) + C
S2       D(I) = A(I+1) + E
    ENDDO
```

- Does it have any dependency?
  Anti or True?
  - **S2 δ^{-1}_{1} S1**



Exercise:

```
    DO I = 1, 100
S1      X(I) = Y(I) + 10
        DO J = 1, 100
S2          B(J) = A(J,N)
            DO K = 1, 100
S3              A(J+1,K)=B(J)+C(J,K)
            ENDDO
S4       Y(I+J) = A(J+1, N)
        ENDDO
    ENDDO
```

	S1	S2	S3	S4
S1				
S2				
S3				
S4				

Exercise

```
      DO I = 1, 100
S1        X(I) = Y(I) + 10
          DO J = 1, 100
S2            B(J) = A(J,N)
              DO K = 1, 100
S3                A(J+1,K)=B(J)+C(J,K)
              ENDDO
S4         Y(I+J) = A(J+1, N)
          ENDDO
      ENDDO
```

	S1	S2	S3	S4
S1	-	-	-	-
S2				
S3				
S4	δ_1			

True dependence from S4 to S1

$I_w + j_w = I_r$

$I_w - I_r = -j_w$

As j_w is always positive Direction is "<"

Exercise

```
    DO I = 1, 100
S1      X(I) = Y(I) + 10
        DO J = 1, 100
S2          B(J) = A(J,N)
            DO K = 1, 100
S3              A(J+1,K)=B(J)+C(J,K)
            ENDDO
S4       Y(I+J) = A(J+1, N)
        ENDDO
    ENDDO
```

	S1	S2	S3	S4
S1	-	-	-	-
S2				
S3				
S4	δ_1			

Exercise

```
    DO I = 1, 100
S1      X(I) = Y(I) + 10
        DO J = 1, 100
S2          B(J) = A(J,N)
            DO K = 1, 100
S3              A(J+1,K)=B(J)+C(J,K)
            ENDDO
S4       Y(I+J) = A(J+1, N)
        ENDDO
    ENDDO
```

	S1	S2	S3	S4
S1	-	-	-	-
S2		$\delta^0{}_1$		
S3				
S4	δ_1			

Exercise

```
    DO I = 1, 100
S1       X(I) = Y(I) + 10
         DO J = 1, 100
S2          B(J) = A(J,N)
            DO K = 1, 100
S3             A(J+1,K)=B(J)+C(J,K)
            ENDDO
S4       Y(I+J) = A(J+1, N)
         ENDDO
    ENDDO
```

	S1	S2	S3	S4
S1	-	-	-	-
S2				
S3				
S4	δ_1			

S2 and S3: dependence via B(J)
I does not occur in either subscript (D.V = *)
We get: Jw = Jr
Direction vectors = (*, =)

Exercise

```
    DO I = 1, 100
S1      X(I) = Y(I) + 10
        DO J = 1, 100
S2          B(J) = A(J,N)
            DO K = 1, 100
S3              A(J+1,K)=B(J)+C(J,K)
            ENDDO
S4       Y(I+J) = A(J+1, N)
        ENDDO
    ENDDO
```

	S1	**S2**	**S3**	**S4**
S1	-	-	-	-
S2	-	δ^0_1	$\delta_1\delta_\infty$	-
S3		δ^{-1}_1		
S4	δ_1			

Exercise

```
    DO I = 1, 100
S1      X(I) = Y(I) + 10
        DO J = 1, 100
S2          B(J) = A(J,N)
            DO K = 1, 100
S3              A(J+1,K)=B(J)+C(J,K)
            ENDDO
S4       Y(I+J) = A(J+1, N)
        ENDDO
    ENDDO
```

	S1	S2	S3	S4
S1	-	-	-	-
S2	-	$\delta^0{}_1$	$\delta_1\delta_\infty\delta^{-1}{}_1$	-
S3	-	$\delta_2\delta_1\delta^{-1}{}_1$		
S4	δ_1			

Exercise

```
    DO I = 1, 100
S1      X(I) = Y(I) + 10
        DO J = 1, 100
S2          B(J) = A(J,N)
            DO K = 1, 100
S3              A(J+1,K)=B(J)+C(J,K)
            ENDDO
S4       Y(I+J) = A(J+1, N)
        ENDDO
    ENDDO
```

	S1	S2	S3	S4
S1	-	-	-	-
S2	-	$\delta^0{}_1$	$\delta_1\delta_\infty\delta^{-1}{}_1$	-
S3	-	$\delta_2\delta_1\delta^{-1}{}_1$	$\delta^0{}_1$	
S4	δ_1			

Exercise

```
      DO I = 1, 100
S1        X(I) = Y(I) + 10
          DO J = 1, 100
S2            B(J) = A(J,N)
              DO K = 1, 100
S3                A(J+1,K)=B(J)+C(J,K)
              ENDDO
S4         Y(I+J) = A(J+1, N)
          ENDDO
      ENDDO
```

	S1	S2	S3	S4
S1	-	-	-	-
S2	-	$\delta^0{}_1$	$\delta_1\delta_\infty\delta^{-1}{}_1$	-
S3	-	$\delta_2\delta_1\delta^{-1}{}_1$	$\delta^0{}_1$	$\delta_1\delta_\infty$
S4	δ_1	-	$\delta^{-1}{}_1$	

Exercise

```
    DO I = 1, 100
S1      X(I) = Y(I) + 10
        DO J = 1, 100
S2          B(J) = A(J,N)
            DO K = 1, 100
S3              A(J+1,K)=B(J)+C(J,K)
            ENDDO
S4       Y(I+J) = A(J+1, N)
        ENDDO
    ENDDO
```

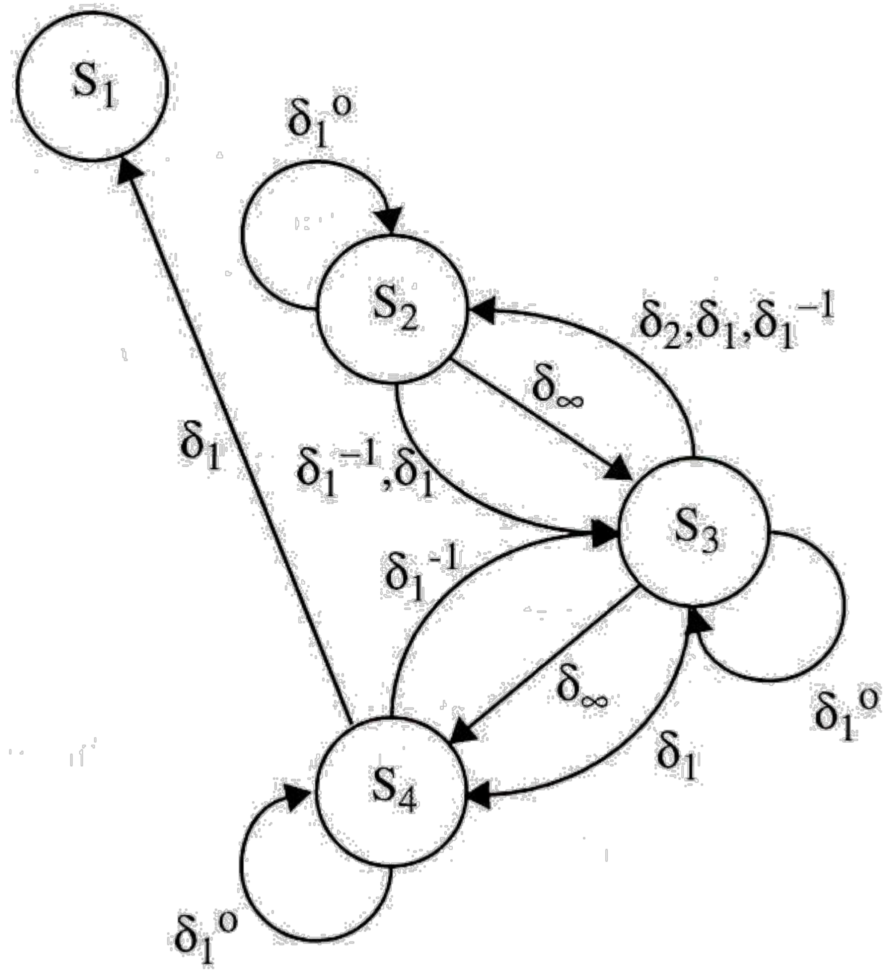
	S1	**S2**	**S3**	**S4**
S1	-	-	-	-
S2	-	$\delta^0{}_1$	$\delta_1\delta_\infty\delta^{-1}{}_1$	-
S3	-	$\delta_2\delta_1\delta^{-1}{}_1$	$\delta^0{}_1$	$\delta_1\delta_\infty$
S4	δ_1	-	$\delta^{-1}{}_1$	$\delta^0{}_1$

Dependency Graph

	S1	S2	S3	S4
S1	-	-	-	-
S2	-	δ^0_1	$\delta_1 \delta_\infty \delta^{-1}_1$	-
S3	-	$\delta_2 \delta_1 \delta^{-1}_1$	δ^0_1	$\delta_1 \delta_\infty$
S4	δ_1	-	δ^{-1}_1	δ^0_1

Dependency Graph

	S1	S2	S3	S4
S1	-	-	-	-
S2	-	$\delta^0{}_1$	$\delta_1\delta_\infty\delta^{-1}{}_1$	-
S3	-	$\delta_2\delta_1\delta^{-1}{}_1$	$\delta^0{}_1$	$\delta_1\delta_\infty$
S4	δ_1	-	$\delta^{-1}{}_1$	$\delta^0{}_1$



Next Lecture

- Algorithm for extracting parallelism from sequential program.

References

- Slides of the book: Randy Allen, Ken Kennedy,Optimizing Compilers for Modern Architectures: A Dependence-based Approach, Morgan Kaufmann, 2001

Questions?

Thank you!